Fiscal Year:	FY 2014	**Task Last Updated:**	FY 04/28/2014
PI Name:	Bouxsein, Mary Ph.D.		
Project Title:	Maintenance of Skeletal Integrity in Spaceflight: Interaction among Partial Weight Bearing, Microgravity and Novel Therapeutics		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	mbouxsei@bidmc.harvard.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	617-667-4594
Organization Name:	Beth Israel Deaconess Medical Center/Harvard Medical School		
PI Address 1:	Department of Orthopedic Surgery		
PI Address 2:	330 Brookline Ave, RN115		
PI Web Page:			
City:	Boston	**State:**	MA
Zip Code:	02215-5400	**Congressional District:**	7
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009 Space Biology-FSB-Animal Physiology: NNH09ZTT003N
Start Date:	04/01/2010	**End Date:**	01/31/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
Contact Email:	jeffrey.d.smith2@nasa.gov		
Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-135 NOTE: Received extension to 1/31/2014 per A. Chu/ARC (Ed., 8/19/13) NOTE: received extension to 9/1/2013 per NSSC (Ed., 4/23/2012)		
Key Personnel Changes/Previous PI:	There have not been any changes to key personnel		
COI Name (Institution):	Rutkove, Seward (Beth Israel Deaconess Medical Center)		
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Task Description:	Despite intensive countermeasures, muscle and bone loss remain among the top medical challenges to extended space missions. Whereas much is known regarding the musculoskeletal responses to microgravity, little is known about the physiological effects of prolonged exposure to partial gravity environments, and whether musculoskeletal deterioration may be mitigated by exposure to partial gravity. To address this knowledge gap, we have developed a novel ground-based model that allows chronic exposure to reduced quadrupedal weightbearing in mice. Using this model will determine whether partial weight bearing attenuates bone and muscle atrophy relative to full hindlimb unloading, and will test the ability of novel spaceflight-compatible therapeutics (e.g., sclerostin antibody and a BMPR1A antagonist) to inhibit skeletal deterioration across the hypogravity spectrum. In addition in a supplement to the original grant, we tested whether sclerostin antibody treatment would inhibit bone loss in microgravity on a 12 day shuttle mission (STS-135). Outcome assessments included in vivo bone density and body composition measurements, along with ex vivo measures of muscle weight, bone microarchitecture, and strength. Serum markers of bone turnover, and quantitatitve histomorphometry allowed us to delineate the mechanisms underlying the observed skeletal changes. In sum, our proposed studies use both a standardized microgravity analog along with a novel partial gravity model and spaceflight exposure to directly address a key element of the Bioastronautics Roadmap--Accelerated Bone Loss and Fracture Risk. Our ultimate objective is to mitigate the risk for space-induced bone loss and bone fracture associated with missions to low earth orbit, the moon, and eventually Mars. Successful completion of these studies will provide strong rationale for testing these pharmacologic countermeasures in human bed-rest studies and then leading to a trial in astronauts traveling to the International Space Station (ISS). Our findings will also provide insight into the prevention and treatment of skeletal fragility due to immobilization, stroke, cerebral palsy, muscular dystrophy, and spinal cord injury.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The current research has profound impact on Earth-based diseases and disorders resulting from insufficient skeletal loading, such as muscular dystrophy, cerebral palsy, stroke, and other neurodegenerative disorders. Our goal is to characterize the amount of skeletal loading necessary to maintain bone and muscle mass, and to determine the efficacy of novel therapeutics in preventing bone and muscle loss. If successful, this would provide novel insight for the rehabilitation of individuals exposed to musculoskeletal unloading or disuse. Our findings indicate that treatment with sclerostin antibody would be likely be effective in preventing disuse-induced bone loss in a variety of clinical conditions, such as spinal cord injury, cerebral palsy, immobilization and stroke.
Task Progress:	In this project we determined the musculoskeletal effects of partial weight bearing in normal mice and assessed the ability of anti-sclerostin antibody (Scl-AbII) and soluble mBMRPR1A–mfc fusion protein to prevent disuse-related bone loss in mice. In addition, in a supplementary grant, we determined the ability of sclerostin antibody to prevent skeletal deterioration during spaceflight by conducting an experiment on the last shuttle mission, STS-135 in July 2011. For our ground-based work, we studied female wild-type, adult mice (C57Bl/6J, 12 wks of age) subjected to partial weight bearing (20% of body weight, PWB20 and 40% of body weight, PWB40, Lunar and Mars loading analogs), 70% of body weight (PWB70) and hindlimb unloading (HLU) for 21 days. Mice were injected with either sclerostin antibody (Scl-AbII) at 25 mg/kg or vehicle twice weekly. In vehicle treated animals we observed declines in bone and muscle that were largely dose-dependent for the partial weight bearing animals, including declines in bone and muscle even at 70% weight bearing. Scl-AbII completely inhibited the trabecular and cortical bone deterioration due to disuse, with bone properties in HLU and partial weight bearing mice treated with Scl-AbII being at or above values of fully loaded, vehicle treated mice. Histomorphometry and serum bone turnover markers indicate that sclerostin antibody induces new bone formation. As expected, there was no effect of Scl-AbII on muscle loss. For some bone outcomes, the efficacy of Scl-AbII was greater in normally loaded animals than in those exposed to hindlimb unloading or partial weightbearing. Altogether, these results indicate: 1) partial weight bearing at loading levels up to 70% of body weight is insufficient to prevent bone and muscle loss; and 2) the potent anabolic activity of Scl-AbII, and its efficacy in abrogating disuse-induced bone deterioration. We were not been able to complete our initial Aim 1 due to our inability to obtain huRANKL mice needed to test the antibody to RANKL. The reason for this is that Amgen was unable to provide the mice due to legal issues related to the intellectual property rights surrounding the development of these mice. Previous work has shown that the soluble murine BMPR1A–fusion protein (mBMPR1A-mFc) binds to BMP2 and BMP4 with high affinity, preventing downstream signaling. Further, treatment of intact and ovariectomized mice with mBMPR1A-mFc leads to increased bone mass, improved bone microarchitecture and strength, via increased bone formation and reduced resorption. Thus, we proposed an alternate to test the ability of a novel anabolic therapy, inhibition of ALK3 receptor signaling via a soluble mBMRPR1A –mfc fusion protein, to inhibit skeletal deterioration with unloading. In this study, we tested the effects of mBMPR1A-mFc on disuse-induced bone loss caused by 21 days of hindlimb unloading (HLU) via tail suspension vs. cage controls (CON). Adult female C57BL/6J mice (12 wks old) were assigned to one of four groups (n=10 each): CON-VEH; CON-mBMPR1A-mFc; HLU-VEH; and HLU-mBMPR1A-mFc. Mice were injected subcutaneously with VEH or mBMPR1A-mFc (4.5 mg/kg, 2x/week). Leg BMD declined in the HLU-VEH group (-5.3 ± 1.3%), whereas it was unchanged in HLU-mBMPR1A-mFc (-0.3 ± 0.9%, $p<0.05$ vs HLU-VEH). Leg BMD increased significantly more in CON-mBMPR1A-mFc than CON-VEH (10.2±0.6% vs 4.4±0.8%). In the femur, trabecular and cortical bone microarchitecture was worse in the HLU-VEH compared to CON-VEH mice, whereas mBMPR1A-mFc treatment led to improved Tb.BV/TV, Tb.Th and midshaft Ct.Th in both the HLU and CON groups ($p<0.05$). HLU-mBMPR1A-mFc mice had 21% greater failure load ($p<0.05$) compared to their VEH-treated counterparts. Dynamic histomorphometry indicated that treatment with mBMPR1A-mFc led to significant increases in mineralizing surface and mineral apposition rate, resulting in a 3.5-fold and 5-fold increase in bone formation rate in the mBMPR1A-mFc -treated CON and HLU animals vs. VEH groups, respectively. mBMPR1A-mFc -treated mice had a similar osteoblast surface but significantly lower osteoclast surface than VEH-treated animals in both the CON and HLU groups. Altogether these findings suggest that treatment with the soluble BMPR1A fusion protein may be useful for maintenance of skeletal integrity in the setting of disuse-induced bone loss. For our spaceflight experiment (Supplementary grant), we successfully completed a 13-day spaceflight with 30 mice on the shuttle STS-135 and 30 mice maintained as ground controls in the orbital environmental simulator at Kennedy Space Center. In addition, tissues from this experiment were shared with over 60 investigators via NASA's biospecimen sharing program. Major findings include: 1) significant bone and muscle loss in the mice exposed to microgravity, and 2) that a single dose of sclerostin antibody was able to prevent skeletal deterioration and induce new bone formation in microgravity. Further, we found that novel muscle measurements, electrical impedance myography, were sensitive to

	the muscle atrophy observed following spaceflight. We also completed a companion ground-based hindlimb unloading (HLU) study, which showed similar results as the spaceflight study. Altogether our results provide compelling rationale to test sclerostin antibody for its ability to prevent bone loss in human studies of disuse, such as bedrest.
Bibliography Type:	Description: (Last Updated: 04/02/2026)
Abstracts for Journals and Proceedings	Lau AG, Ortega AM, Bouxsein ML, Bateman TA, Hanson A, Livingston EW, Pruitt T, Smith CT, Derosa AB, Lai EW, Bowman LC, Stodieck LS, Ellman R, Spatz J, Warmington K, Tan HL, Hill D, Maurya S, Cureton A, Lotinun S, Paszty C, Ferguson VL. "How Spaceflight and Sclerostin Antibody Treatment Affect the Mechanical and Microstructural Properties of Bone in Mice Flown on STS-135 using Finite Element Modeling. using Finite Element Modeling" 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. Poster #1146. , Feb-2013
Abstracts for Journals and Proceedings	Sullivan LK, Ortega AM, Bateman TA, Hanson A, Livingston EW, Pruitt T, Smith CT, de Rosa AB, Lai EW, Stodieck LS, Bowman LC, Ellman R, Spatz J, Warmington K, Tan HL, Hill D, Paietta R, Cureton A, Paszty C, Bouxsein ML, Ferguson VL, Lau AG. "Effects of Spaceflight and Sclerostin Antibody on Femoral Neck Strength Estimated by Finite Element Analysis in Mice Flown on Space Shuttle Flight STS-135." 35th Annual Meeting of the American Society for Bone and Mineral Research, Baltimore, Maryland, October 4-7, 2013. J Bone Miner Res 2013;28(Suppl 1). Poster #: MO0058. Available at: http://www.asbmr.org/education/AbstractDetail?aid=4473050e-ab51-456c-8b95-25123dd743ab ; Accessed April 29, 2014. , Oct-2013
Abstracts for Journals and Proceedings	Marshall EA, Sullivan LK, Livingston EW, Hanson A, Pruitt T, Stodieck L, Bowman LC, Ellman R, Spatz J, Paszty C, Bouxsein ML, Ferguson VL, Bateman TA, Lau AG. "Effects of Spaceflight on the 'Non-Weight Bearing' L5 Vertebra Measured by Finite Element Analysis in Mice Flown on Space Shuttle Flight STS-135." 35th Annual Meeting of the American Society for Bone and Mineral Research, Baltimore, Maryland, October 4-7, 2013. J Bone Miner Res 2013;28(Suppl 1). Poster #SA04. Available at: http://www.asbmr.org/education/AbstractDetail?aid=a4a70f3e-6b61-4f42-8c69-07b695b1b0e5 ; Accessed April 29, 2014. , Oct-2014
Abstracts for Journals and Proceedings	Lau AG, Sullivan LK, Marshall EA, Smith CT, Livingston EW, de Rosa AB, Lai EW, Bowman LC, Stodiek LS, Hanson A, Pruitt T, Ellman R, Spatz J, Paszty C, Ferguson VL, Bouxsein ML, Bateman TA. "Computational Modeling to Determine Functional Changes in Bone Stiffness and Structural Efficiency in Mice Flown on Space Shuttle STS-135." 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. Poster Presentation #3199. http://www.hou.usra.edu/meetings/hrp2014/pdf/3199.pdf , Feb-2014
Abstracts for Journals and Proceedings	Sullivan LK, Ortega AM, Cureton A, Paietta R, Livingston E, de Rosa AB, Lai EW, Smith CT, Bowman LC, Stodieck LS, Hanson A, Pruitt T, Ellman R, Spatz J, Paszty C, Bouxsein ML, Ferguson VL, Bateman TA, Lau AG. "Effects of Microgravity and Sclerostin Antibody on Finite Element Modeling Stiffness of Mouse Femoral Neck." 29th Meeting of the American Society for Gravitational and Space Research (ASGSR), Orlando, FL, November 3-8, 2013. 29th Meeting of the American Society for Gravitational and Space Research (ASGSR), Orlando, FL, November 3-8, 2013. Program and abstracts. Presentation #C5.3 and Poster Presentation #: SP.11. p. 53-54; p. 134. https://asgsr.org/images/stories/pdf/Complete_Program_FINAL_2013_Meeting.pdf , Nov-2013
Abstracts for Journals and Proceedings	Lau AG, Smith CT, Livingston EW, de Rosa AB, Lai EW, Bowman LC, Stodiek LS, Hanson A, Pruitt T, Ellman R, Spatz J, Paszty C, Ferguson VL, Bouxsein ML, Bateman TA. "Shuttle Flight STS-135: Microgravity and Sclerostin on Proximal Tibia Stiffness and Structural Efficiency in Mice." 29th Meeting of the American Society for Gravitational and Space Research (ASGSR), Orlando, FL, November 3-8, 2013. Oral Presentation #C5.1. 29th Meeting of the American Society for Gravitational and Space Research (ASGSR), Orlando, FL, November 3-8, 2013. p. 53. https://asgsr.org/images/stories/pdf/Complete_Program_FINAL_2013_Meeting.pdf , Nov-2013
Abstracts for Journals and Proceedings	Spatz JM, Ayturk UM, van Vliet M, Ellman R, Warman ML, Bouxsein ML. "RnaSeq analysis of gene expression in murine hind limb unloading disuse induced bone loss." 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. Poster. 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. http://www.hou.usra.edu/meetings/hrp2014/pdf/3130.pdf , Feb-2014
Abstracts for Journals and Proceedings	Ellman RE, Conlon C, van Vliet M, Spatz JM, Louis L, Willey JS, Stodieck LS, Bateman TA, Ferguson VL, Bouxsein ML. "Evaluation of the hindlimb unloading model in mice as an analog for spaceflight effects on the musculoskeletal system." 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. Poster presentation. http://www.hou.usra.edu/meetings/hrp2014/pdf/3262.pdf , Feb-2014
Abstracts for Journals and Proceedings	Sullivan LK, Livingston EW, Ellman R, Spatz J, Stodieck LS, Bouxsein ML, Ferguson VL, Bateman TA, Lau AG. "Effects of Hind Limb Unloading and Sclerostin Antibody on Femoral Neck Strength Estimated by Finite Element Analysis." 36th Annual Meeting of the American Society for Bone and Mineral Research, Houston, Texas, September 12-15, 2014. 36th Annual Meeting of the American Society for Bone and Mineral Research, Houston, Texas, September 12-15, 2014. Submitted to J Bone Miner Res 29(Suppl 1): in press, as of April 2014. , Apr-2014
Abstracts for Journals and Proceedings	Comte T, Lau AG, Livingston EW, Ellman R, Spatz J, Stodieck LS, Bouxsein ML, Ferguson VL, Bateman TA. "Proximal Tibia Bone Stiffness and Structural Efficiency in Spaceflight and Hind Limb Unloading with a Sclerostin Antibody Countermeasure." 36th Annual Meeting of the American Society for Bone and Mineral Research, Houston, Texas, September 12-15, 2014. 36th Annual Meeting of the American Society for Bone and Mineral Research, Houston, Texas, September 12-15, 2014. Submitted to J Bone Miner Res 29(Suppl 1): in press, as of April 2014. , Apr-2014

Abstracts for Journals and Proceedings	Lau AG, Ortega AM, Bouxsein ML, Bateman TA, Hanson A, Livingston EW, Pruitt T, Smith CT, Derosa AB, Lai EW, Bowman LC, Stodieck LS, Ellman R, Spatz J, Warmington K, Tan HL, Hill D, Maurya S, Cureton A, Lotinun S, Paszty C, Ferguson VL. "Effects of spaceflight and a sclerostin antibody countermeasure on the mechanical properties of bone in mice." 2012 Annual Meeting of the American Society for Bone and Mineral Research, Minneapolis, Minnesota, October 12-15, 2012. Poster Presentation. J Bone Miner Res 2012;27(Suppl 1). http://www.asbmr.org/Meetings/AnnualMeeting/AbstractDetail.aspx?aid=f341a654-1efa-4f14-b89a-4a2d536f88d7 ; Accessed April 29, 2014. , Oct-2012
Abstracts for Journals and Proceedings	Ellman RE, Conlon C, van Vliet M, Spatz J, Louis L, Willey J, Stodieck L, Bateman T, Ferguson V, Bouxsein ML. "Evaluation of the tail suspension model in mice as an analog for spaceflight effects on the musculoskeletal system" 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, November 27-December 2, 2012. Oral presentation. 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, November 27-December 2, 2012. Program and abstracts, p. 87. https://asgsr.org/images/stories/pdf/2012_AnnualMeeting_Program.pdf , Nov-2012
Articles in Peer-reviewed Journals	Jonscher KR, Alfonso-Garcia A, Suhalim JL, Orlicky DJ, Potma EO, Ferguson VL, Bouxsein ML, Bateman TA, Stodieck LS, Levi M, Friedman JE, Gridley DS, Pecaut MJ. "Spaceflight activates lipotoxic pathways in mouse liver." PLoS One. 2016 Apr 20;11(4):e0152877. eCollection 2016. http://dx.doi.org/10.1371/journal.pone.0152877 ; PubMed PMID: 27097220; PubMed Central PMCID: PMC4838331 Erratum in: PLoS One. 2016;11(5):e0155282. http://dx.doi.org/10.1371/journal.pone.0155282 , Apr-2016
Articles in Peer-reviewed Journals	Spatz JM, Ellman R, Cloutier AM, Louis L, van Vliet M, Dwyer D, Stolina M, Ke HZ, Bouxsein ML. "Sclerostin antibody inhibits skeletal deterioration in mice exposed to partial weight-bearing." Life Sci Space Res. 2017 Feb;12:32-8. http://dx.doi.org/10.1016/j.lssr.2017.01.001 , Feb-2017
Articles in Peer-reviewed Journals	Ko FC, Van Vliet M, Ellman R, Grasso D, Brooks DJ, Spatz JM, Conlon C, Aguirre JI, Wronski TJ, Bouxsein ML. "Treatment with a soluble bone morphogenetic protein type 1A receptor (BMPR1A) fusion protein increases bone mass and bone formation in mice subjected to hindlimb unloading." JBMR Plus. 2017 Oct;1(2):66-72. eCollection 2017 Oct. https://doi.org/10.1002/jbm4.10012 ; PubMed PMID: 30283882; PubMed Central PMCID: PMC6124165 , Oct-2017
Articles in Peer-reviewed Journals	Coulombe JC, Senwar B, Ferguson VL. "Spaceflight-induced bone tissue changes that affect bone quality and increase fracture risk." Curr Osteoporos Rep. 2020 Feb;18(1):1-12. Review. https://doi.org/10.1007/s11914-019-00540-y ; PubMed PMID: 31897866 , Feb-2020
Articles in Peer-reviewed Journals	Spatz JM, Ko FC, Ayturk UM, Warman ML, Bouxsein ML. "RNAseq and RNA molecular barcoding reveal differential gene expression in cortical bone following hindlimb unloading in female mice." PLoS One. 2021 Oct 12;16(10):e0250715. https://doi.org/10.1371/journal.pone.0250715 ; PMID: 34637435; PMCID: PMC8509868 , Oct-2021
Articles in Peer-reviewed Journals	Ellman R, Spatz J, Cloutier A, Palme R, Christiansen B, Bouxsein ML. "Partial reductions in mechanical loading yield proportional changes in bone density, bone architecture, and muscle mass." J Bone Miner Res. 2013 Apr;28(4):875-85. http://dx.doi.org/10.1002/jbmr.1814 ; PubMed PMID: 23165526 , Apr-2013
Articles in Peer-reviewed Journals	Spatz JM, Ellman R, Cloutier AM, Louis L, van Vliet M, Suva LJ, Dwyer D, Stolina M, Ke HZ, Bouxsein ML. "Sclerostin antibody inhibits skeletal deterioration due to reduced mechanical loading." J Bone Miner Res. 2013 Apr;28(4):865-74. http://dx.doi.org/10.1002/jbmr.1807 ; PMID: 23109229 , Apr-2013
Articles in Peer-reviewed Journals	Sung M, Li J, Spieker AJ, Spatz J, Ellman R, Ferguson VL, Bateman TA, Rosen GD, Bouxsein M, Rutkove SB. "Spaceflight and hind limb unloading induce similar changes in electrical impedance characteristics of mouse gastrocnemius muscle." J Musculoskelet Neuronal Interact. 2013 Dec;13(4):405-11. PubMed PMID: 24292610 ; http://www.ismni.org/jmni/pdf/54/02SUNG.pdf , Dec-2013
Articles in Peer-reviewed Journals	Ellman R, Grasso DJ, van Vliet M, Brooks DJ, Spatz JM, Conlon C, Bouxsein ML. "Combined effects of botulinum toxin injection and hind limb unloading on bone and muscle." Calcif Tissue Int. 2014 Mar;94(3):327-37. Epub 2013 Nov 17. http://dx.doi.org/10.1007/s00223-013-9814-7 ; PubMed PMID: 24240478; PubMed Central PMCID: PMC3921683 , Mar-2014